GRIDZILLA

A grid signal for home batteries, cars and heat pumps.

Mathew Stafurik · Peter Mikeška · Matthew · Tedi

Grid-Serving Storage · Energy Bridge Vienna 2026

Europe's largest battery is in people's basements.

The grid can use none of it.

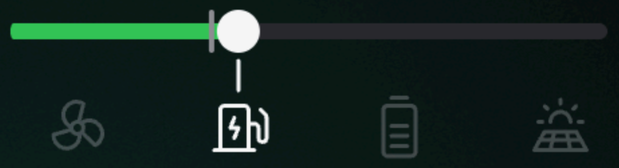
Slovakia



0 signals

from the grid operator
to home batteries

Germany



4.2 kW

the operator may dim
one new charger to this

Ukraine



3 GWh

of home batteries, idle
through the blackouts

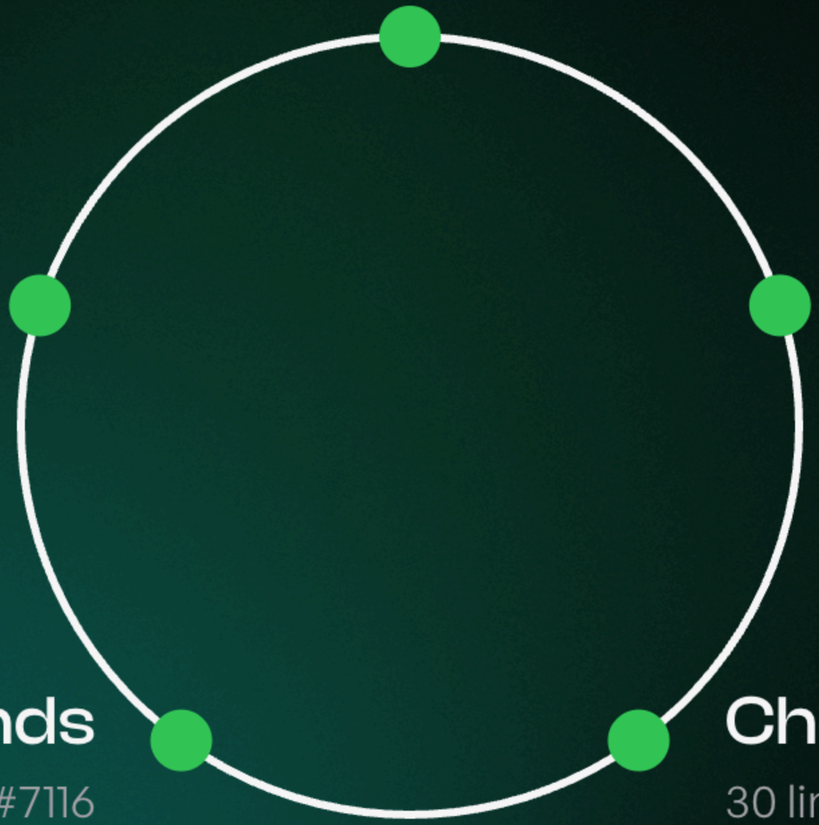
The storage exists. The signal does not.



Battery first. Car second. Hot water third.

The operator asks. The house decides. The owner can always say no.

The loop, every 15 minutes.



Gathers

72 transformers, the weather, the plan, every reply

Thinks

The AI writes the message in 1.9 s

Checks

30 lines of plain rules, not AI

Signs and sends

The platform signs message #7116

Houses reply

271 homes answer, all signed

Six words. Signed.

SUPPORT

Evening peak, or the grid short of power

“Take at most 3 kW from the street.”

ABSORB

Noon sun flooding the street

“Send back at most 2 kW.”

PREPARE

A warning: storm, planned outage, attack

“Be 90 % full by 17:00.”

STRETCH

The street is dark, power back at a known time

“Run on your own, send nothing back, make it last.”

RECOVER

The transformer is replaced

“Refill slowly, not all at once.”

RELEASE

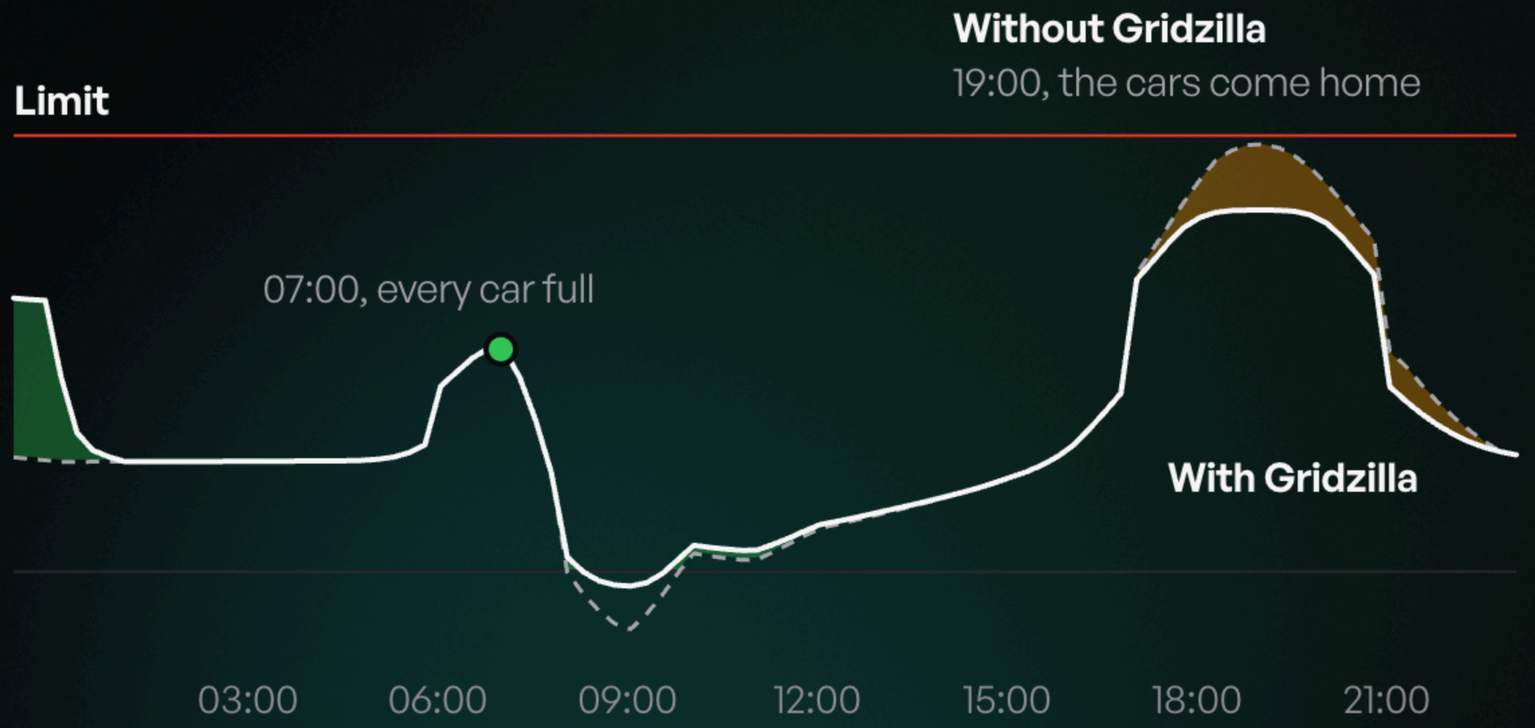
The event is over

“You are on your own again.”

Plus a 24-hour envelope per street, renewed every two hours. Every message Ed25519-signed with a time window, ramps randomised so a street never steps in sync, the owner's limits always win, and everything reverts when the heartbeat stops.

The monster eats peaks.

Every street, every evening.



83%

with Gridzilla, tonight's plan down from 98 % without it

The next 24 hours for Vienna: 2,168 homes, cars and heat pumps, renewed every two hours from the real weather. Live.

Saved tonight

4.5 MW
off the city's evening peak

3.2 MW
kept off the grid right now

32 of 32
streets kept inside their limit

10,419 kWh
car charging moved into the night

What we did

15
signals signed today

648
houses following a signal now

Essling

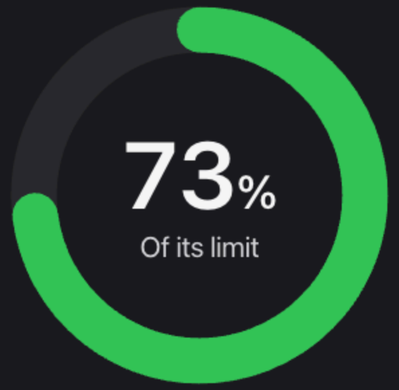
Donaustadt street transformer · limit 400 kW

Peak coming tomorrow at 18:53: 119 % of the limit. Gridzilla is moving 117 kW to keep Essling at 90 %.

Weather Wien-Liesing: 6.8 °C, feels 3.3 °C · cloud 100 % · sun 161 W/m² · wind 9.7 km/h · no official warnings · 9 °C at 04:00 (feels 5 °C), so heat pumps run hard; gusts up to 63 km/h around 23:00.

With Gridzilla (measured) · Without Gridzilla (what the houses would draw) · Peak eaten · Charging moved · The plan · Limit · Signal until **01:38**





What just happened

All events

23:58 Use your battery on substation Schwarzenbergplatz ended: its time window ran out
23:18 Use your battery on substation Wien Sued ended: its time window ran out
23:18 AI proposed Use your battery on feeder Donaustadt 4 · guard passed · signed #7116 · 96 of 96 houses applied

Now

next DeepSeek review in 10 s · autonomous

Sees
120 % at 22:58
Suessenbrunn

Thinks
DeepSeek 1.9 s
Use your battery

Checks
Guard passed
inside every safety rule

Acts
#7116 signed
at most 3 kW until 01:18, 96 homes

Signals running

Use batteries SUPPORT on Donaustadt — **1:40 left**
signed #7116 · take at most 3 kW · 8 time slots — limit 8.7–8.8 kW — Release

Use batteries SUPPORT on Leobendorf 62 — **1:08 left**
signed #7091 · take at most 3 kW · 8 time slots — limit 8.2–8.7 kW — Release

and 2 more running (all of them in Operations)

Blatnica · real plant · live

live · updated 14 s ago · Cerbo GX + Vitocal via Gridzilla agents

+0.08 kW
grid import

32 %
battery SoC · 15 kWh · owner floor 20 %

0.0 kW
EV charger · 11 kW station · idle

0.0 kW
heat pump · tank 50.6 °C → 50 °C · room → 23 °C · floor flow 34 °C



grid kW, + import − export, last 30 min

owner mode · last ack reverted (window_end) · signature ✓

Wien Sued · Wien Nord · Wien West · Wien Ost · Kendlerstrasse · Wien Mitte · Bisamberg · Schwarzenbergplatz

Our family home in Blatnica

Victron battery
32%

Home at the meter
0.3kW

Car charger
0.0kW

Viessmann heat pump
0.0kW

Live, on the same signed message path

Signed
by the platform, one message

Checked
by the plant, signature first

Followed
by the battery and heat pump

Real end to end. The planning runs on the fleet.
Paid by the meter: a flat yearly fee, crisis prices only when the meter proves the limit was kept.

A simulated Vienna

Two thousand houses, the same cars with and without Gridzilla, realistic charging.



72 street transformers in a simulated Vienna, live

72 inside the limit · 0 near it · 0 over

−36%
one street
Atzgersdorf, 511 to 325 kW

−5MW
Vienna
two thousand houses, 31.2 to 26.2 MW

0kW
a street
where no car charged

Measured on the simulated fleet, 24 Oct 2026, with the heat pumps shedding floor heating as well

It eats peaks for breakfast.

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